CARDIOVASCULAR DISEASE
RISK PREDICTION

A Machine Learning Classification Study

Ensemble Methods, Threshold Optimisation & Clinical Decision Support

Dataset: ~70,000 patient records · 11 Features · Binary Classification
Models Evaluated: 11 · Final Model: Random Forest Classifier

Best CV Precision: ~75–76% · OOB Score: ~73% · Recall @ 0.30 threshold: 89%

1. Project Overview

This project builds and evaluates a suite of machine learning classifiers to predict the presence or absence of cardiovascular disease (CVD) in a patient population of approximately 70,000 records. The goal is both predictive accuracy and clinical utility — specifically minimising missed disease cases (false negatives), which carry greater patient risk than false positives.

Eleven models were trained, tuned, and compared using stratified cross-validation. The final model — a Random Forest Classifier — was selected for its stable precision (~75–76%), controlled variance, and interpretable feature importance. Threshold analysis was then applied to calibrate the model for clinical deployment scenarios.

2. Dataset Description

The dataset is sourced from Kaggle (Cardiovascular Disease Dataset) and contains 70,000 records at intake. Each record describes a single patient across three categories of features: objective (measured), examination (clinically assessed), and subjective (self-reported).

Feature	Category	Type	Description/Notes
age_yr	Objective	Continuous	Age converted from days to years
gender	Objective	Binary	0 = Female, 1 = Male (inferred from mean height)
BMI_kg/m2	**Engineered**	**Continuous**	**Derived from height & weight; height & weight dropped post-calculation**
ap_hi	Examination	Continuous	Systolic blood pressure (mmHg) — heavily cleaned
ap_lo	Examination	Continuous	Diastolic blood pressure (mmHg) — heavily cleaned
cholesterol	Examination	Ordinal	1: Normal, 2: Above normal, 3: Well above normal
gluc	Examination	Ordinal	1: Normal, 2: Above normal, 3: Well above normal
smoke	Subjective	Binary	0 = Non-smoker, 1 = Smoker
alco	Subjective	Binary	0 = No alcohol intake, 1 = Alcohol intake
active	Subjective	Binary	0 = Not active, 1 = Physically active
cardio	**Target**	**Binary**	**0 = No CVD, 1 = CVD present**

Key data engineering steps: age was converted from days to years; height and weight were used to compute BMI (kg/m²) and then dropped; the unnamed ID column was removed. The target variable (cardio) is approximately balanced (~50/50 split), confirming no sampling adjustment was necessary.

Gender Inference Note: The dataset did not document which encoded value represented male or female. By computing the mean height of each gender group (0 = 161 cm, 1 = 169 cm) and applying the known biological tendency for males to be taller, it was inferred that 1 = Male and 0 = Female.

3. Data Cleaning & Quality Treatment

3.1 Blood Pressure Cleaning Pipeline

Systolic (ap_hi) and diastolic (ap_lo) blood pressure values were the most problematic features, containing typos, sign errors, and physiologically impossible values. A sequential rule-based cleaning pipeline was applied:

Condition Detected	Action Taken
ap_lo ≥ 1000 & ap_lo ≥ ap_hi	Typo (extra zero) — divide offending rows by 10
ap_lo = 0 or ap_hi = 0	Physiologically impossible — rows deleted
ap_hi < 0 or ap_lo < 0	Sign error — convert to positive; row 60106 (irrecoverable) deleted
ap_hi > 10,000 or ap_lo > 10,000	Divide ap_hi by 100 (typo pattern confirmed)
ap_hi > 1,000 or ap_lo > 1,000	Divide ap_hi by 10; residual outliers deleted
ap_hi > 300 or ap_lo > 300	Beyond physiological range — rows deleted

Validated physiological ranges: ap_hi in 90–200 mmHg; ap_lo in 60–130 mmHg; mandatory constraint: ap_lo < ap_hi at all times.

```
# Step 1 — Fix typo: ap_lo >= 1000 and ap_lo >= ap_hi
df[(df['ap_lo'] >= df['ap_hi']) & (df['ap_lo'] >= 1000)] /= 10

# Step 2 — Remove zero blood pressure rows
df = df[((df['ap_lo'] != 0) & (df['ap_hi'] != 0))]

# Step 3 — Flip negative values to positive; remove irrecoverable row
df = df.drop(index=[60106])
idx_neg = list(df[(df['ap_hi'] < 0) | (df['ap_lo'] < 0)].index)
df.loc[idx_neg, 'ap_hi'] = -df.iloc[idx_neg, 2]

# Step 4-6 — Handle extreme large values and physiological ceiling (>300)
df = df.drop(index=df[(df['ap_hi'] > 300) | (df['ap_lo'] > 300)].index)
```

3.2 BMI Validation

After computing BMI from height and weight, values outside the range 10–100 kg/m² were removed as physiologically implausible. Although BMI was computed early in the pipeline, the validation logic is equivalent regardless of step order.

```
df = df.drop(index=df[(df['BMI_kg/m2'] < 10) | (df['BMI_kg/m2'] > 100)].index)
```

3.3 Missing Values & Duplicates

After all transformations, 7 missing values were identified in ap_hi — a negligible count at random — and removed via dropna(). Duplicate rows were also removed. The final cleaned dataset contains approximately 68,000+ records ready for modelling.

4. Machine Learning Modelling

4.1 Train-Test Strategy

The dataset was split 80/20 (train/test) using stratified splitting to preserve the class distribution in both subsets. Stratification prevents sampling bias and ensures fair evaluation across the balanced binary target.

```
X_train, X_test, y_train, y_test = train_test_split(
    df.drop('cardio', axis=1), df['cardio'],
    test_size=0.2, random_state=43, stratify=df['cardio']
)
```

All models were evaluated primarily on cross-validated precision. Accuracy, recall, and F1-score were tracked as secondary metrics.

4.2 Model Inventory & Results

Eleven classification models were built and evaluated. The table below summarises all experiments.

Model	Configuration	Acc.	Prec.	Recall	Best
Logistic Regression	GridSearchCV (C, l1_ratio)	~72%	–	–	
LR + Polynomial Features	Poly(2) + Scaler + GridSearchCV	~72%	–	–	
Decision Tree	GridSearchCV (depth 8, leaf 30)	~73%	~71%	~78%	
Random Forest	**GridSearchCV (depth 10, feat 4)**	**~75%**	**~77%**	**~70%**	
Categorical Naïve Bayes	Default	~72%	~76%	–	

SVM (RBF)	C=10, γ=0.1 — ~25 min/fit	~82%*	~72%†	–
Gradient Boosting	depth 5, lr 0.3, n=300	~79%	77–81%	–
KNN	k=41–107, StandardScaler	~73%	~76%	~70%
Bagging (SVM base)	10 estimators, 60% samples	~78%	–	–
Voting Classifier	Soft voting, 8 models	~73%	–	–
PCA (4 components) + SVM	PCA → SVM pipeline	~72%	–	–

* SVM single-split accuracy; † SVM 5-fold cross-validated precision. Training time ~25 min/fit made SVM impractical for tuning at scale.

4.3 Model Descriptions

Logistic Regression (Models 1 & 2)

A standard Elastic Net LR was tuned with GridSearchCV over C and l1_ratio. Polynomial feature expansion (degree 2) was then added — neither variant exceeded ~72% accuracy, confirming linear decision boundaries are insufficient for this feature space.

Decision Tree (Model 3)

Tuned with GridSearchCV: max_depth=8, max_features=6, min_samples_leaf=30, min_samples_split=70. Slightly improved recall (~78%) but showed higher variance than ensemble methods.

Random Forest (Model 4)

Two rounds of GridSearchCV — final params: max_depth=10, max_features=4, min_samples_leaf=45, min_samples_split=30, n_estimators=200. Achieved ~75–76% precision with OOB score ~73%.

Categorical Naïve Bayes (Model 5)

Applied directly to the feature set without scaling. Precision ~76% but cross-validated accuracy ~72%. Limited by the independence assumption.

SVM RBF (Model 6)

SVC with C=10, γ=0.1 showed 82% single-split accuracy but only ~72% cross-validated precision. Training time ~25 minutes per fit made hyperparameter tuning computationally impractical.

Gradient Boosting (Model 7)

GradientBoostingClassifier with n_estimators=300, max_depth=5, learning_rate=0.3, subsample=0.7. Achieved ~79% train accuracy; cross-validated precision dropped to ~72%, indicating variance sensitivity.

KNN (Model 8)

Three rounds of GridSearchCV expanding k from 5 to 171. StandardScaler applied to continuous features via ColumnTransformer. Best k found in range 41–107, uniform weighting. Precision ~76%, accuracy ~73%.

Bagging (SVM base) & Voting Classifier (Models 9 & 10)

A BaggingClassifier with 10 SVM estimators on 60% samples achieved ~78% accuracy. A VotingClassifier (soft voting) over 8 models achieved ~73% — neither significantly improved upon Random Forest.

PCA + SVM (Model 11)

PCA retained 4 components (sufficient to explain ~90% variance) followed by SVM. Accuracy ~72% — dimensionality reduction did not improve separability for this dataset.

5. Feature Transformation Experiment: BP Discretization

To assess whether clinically meaningful binning improves model performance, systolic and diastolic blood pressure values were discretized into ordinal categories based on standard hypertension guidelines.

5.1 Discretization Scheme

Systolic Blood Pressure (ap_hi):

Systolic Range (mmHg)	Clinical Category	Encoded Value
< 120	Normal	0
120–129	Elevated	1
130–139	Hypertension Stage 1	2
140–179	Hypertension Stage 2	3
≥ 180	Hypertensive Crisis	4

Diastolic Blood Pressure (ap_lo):

Diastolic Range (mmHg)	Clinical Category	Encoded Value
< 80	Normal	0
80–89	Hypertension Stage 1	2
90–119	Hypertension Stage 2	3
≥ 120	Hypertensive Crisis	4

Additional rule: If systolic is 120–129 mmHg AND diastolic < 80 mmHg → Elevated (1).

5.2 Cross-Validated Precision Comparison

The three strongest models (Random Forest, Gradient Boosting, XGBoost) were evaluated on both the original continuous BP data and the discretized version:

Model	Original Data CV Precision	Discretized BP CV Precision
Random Forest	75.73%	75.58%
Gradient Boosting	75.21%	75.18%
XGBoost	73.61%	74.01%

Finding: The differences are negligible (<0.5% across all models). Tree-based models inherently learn optimal split thresholds during training, making manual clinical binning unnecessary for predictive performance.

6. Final Model — Random Forest Classifier

6.1 Model Configuration

The final model configuration, selected after two rounds of GridSearchCV:

```
from sklearn.ensemble import RandomForestClassifier

rf1 = RandomForestClassifier(
    max_depth        = 10,
    max_features     = 4,
    min_samples_leaf = 45,
    min_samples_split= 30,
    n_estimators     = 200,
    oob_score        = True,
    random_state     = 45
)
rf1.fit(X_train, y_train)
# OOB Score: ~0.73
```

6.2 Model Stability & Generalisation

- **Cross-validated precision: ~75–76%**
- Out-of-Bag (OOB) score: ~73% — independent internal validation without a separate hold-out set.
- Train-test gap: minimal — no evidence of overfitting or underfitting.
- Performance consistency confirmed across 5-fold stratified cross-validation.

6.3 Why Random Forest Over Others

Criterion	Justification
Performance	Highest and most stable cross-validated precision (~75–76%)
Variance	Lower variance than Gradient Boosting — cross-val didn't drop to 72%
Compute	Far faster to train/tune than SVM (~25 min/fit) or Gradient Boosting
Interpretability	Feature importance scores directly quantify each variable's contribution

7. Feature Importance Analysis

The Random Forest's feature_importances_ attribute provides a ranked contribution of each input variable to the ensemble's decision-making:

Rank	Feature	Importance	Business Insight
1st	**ap_hi**	**>50% importance**	**Dominant predictor — captures hypertension signal directly**
2nd	ap_lo	~20%	Diastolic pressure — complements systolic reading
3rd	age_yr	~10%	Age is a well-established CVD risk factor
4th	cholesterol	~8%	Moderate impact; ordinal encoding preserves ranking
5th	BMI_kg/m2	~5%	Engineered feature; lower than expected relative to BP
Low	smoke/alco/active	~1–2% each	Lifestyle factors — minimal signal in this feature set

Blood pressure features together contribute approximately 70% of total feature importance. The dominance of ap_hi (>50%) aligns with clinical knowledge: hypertension is the leading modifiable risk factor for cardiovascular disease.

Clinical Insight: The low importance of lifestyle variables (smoke, alco, active) likely reflects that this dataset captures existing conditions rather than long-term lifestyle trajectories. These variables may carry more signal in longitudinal cohort studies.

8. Test Set Performance & Threshold Analysis

8.1 Default Threshold (0.50) — Classification Report

Results on held-out test data (~20% of cleaned dataset) at the default decision threshold of 0.5:

Class	Precision	Recall	F1-Score
0 – No Disease	0.71	0.76	0.73
1 – Disease Present	**0.75**	**0.69**	**0.72**

The model correctly identifies 76% of true non-disease cases (Class 0 recall) but misses 31% of actual disease cases (Class 1 recall = 69%). In a cardiovascular screening context, this 31% miss rate is the critical concern.

> *Medical Priority: False Negatives (missed disease cases) are more dangerous than False Positives in cardiovascular prediction. A patient told they are healthy when they have CVD may delay life-saving treatment.*

8.2 ROC-AUC & Optimal Threshold

The ROC-AUC curve was computed using predict_proba scores. The Youden's J statistic (maximising TPR − FPR) was applied to identify the optimal threshold:

```
fpr, tpr, thresholds = roc_curve(y_test, rf1.predict_proba(X_test)[:, 1])
best_thresh = thresholds[(tpr - fpr).argmax()]
# Result: best_thresh ≈ 0.50  (same as model default)
```

The Youden statistic confirmed that the model's default threshold of 0.50 is already mathematically optimal on this dataset. However, for clinical deployment, a lower threshold may be preferable to prioritise recall.

8.3 Threshold Sensitivity Analysis

Three lower thresholds were evaluated to assess the precision-recall trade-off for clinical applications:

Threshold	Precision	Recall (Disease)	Use Case
0.50 (default)	75%	69%	Balanced — default behaviour
0.40	69%	80%	More recall, fewer missed cases
0.35	66%	85%	High recall — good for broad screening

0.30	**62%**	**89%**	**Maximum recall — population screening**

As the threshold decreases, recall (disease detection rate) increases at the cost of precision (false alarm rate). The optimal threshold depends on the deployment context:

- Threshold 0.30–0.35: Preferred for population-level screening programmes — minimise missed cases.
- Threshold 0.40–0.50: Appropriate for resource-constrained diagnostic triage — reduce unnecessary follow-up.

9. Conclusions & Recommendations

This project successfully developed and validated a machine learning classifier for cardiovascular disease risk prediction on a dataset of 70,000 patient records. Key conclusions:

- **Final model: Random Forest Classifier — CV Precision ~75–76%, OOB Score ~73%, Test Accuracy ~74–76%.**
- 11 models were evaluated; linear models plateaued at ~72%; ensemble methods achieved ~75–79% before cross-validation variance reduction.
- **Systolic blood pressure (ap_hi) is the single largest predictive feature, contributing >50% of model importance.**
- Combined blood pressure features (ap_hi + ap_lo) account for ~70% of model decision-making — validating clinical domain knowledge.
- BP discretization into clinical categories did not improve model performance over continuous values — tree-based models learn optimal splits automatically.
- At threshold 0.30, recall for disease class reaches 89% — suitable for broad population screening with acceptable false positive rate.

Master's-Level Summary: Model performance converged within ~73–78% across all strong non-linear classifiers. This plateau reflects inherent feature-set limitations — not model choice. The dataset lacks detailed biomarkers, genetic indicators, and longitudinal history. Further accuracy gains require richer data, not more complex algorithms. The correct clinical decision is to deploy Random Forest at a threshold of 0.30–0.35 for screening, escalating flagged patients to diagnostic workup.

10. Limitations

Feature Set Constraints

The dataset relies on demographic and basic clinical indicators. Biomarkers (troponin, BNP), imaging results, genetic data, and medication history — all clinically significant — are absent and likely constrain achievable accuracy.

Single Dataset Evaluation

All training and evaluation was performed on a single dataset from one source. External validation on independent hospital populations would be required to assess real-world generalisability before clinical deployment.

Lifestyle Variable Limitation

Smoking, alcohol, and physical activity were self-reported and binary-encoded. Self-report bias and crude encoding likely understate the true contribution of these risk factors.

Tree Importance Bias

Random Forest feature importance scores can be biased towards continuous variables with high cardinality (e.g., ap_hi, age). SHAP (SHapley Additive exPlanations) values would provide more reliable individual-level attribution.

Threshold Deployment Dependency

The optimal decision threshold varies by clinical setting. A universal threshold is inappropriate — the operating threshold must be calibrated to the specific population prevalence and resource context of the deployment environment.